

# BLADE: Bilevel Low-rank Augmented-Lagrangian Driven Erasure for LLM Unlearning

Anonymous ACL submission

## Abstract

Large language models memorize training data, creating legal, privacy, and safety risks that demand targeted knowledge removal without full retraining. Current unlearning methods pair unbounded forget objectives—which destroy model coherence—with fixed-weight loss balancing that cannot track the shifting difficulty of retain preservation during training.

We propose BLADE, a bilevel optimization framework for LLM unlearning. The forget objective is *clamped entropy*: per-token output entropy is maximized only up to a fraction  $\tau$  of maximum entropy  $H_{\max} = \log V$ , after which the gradient is exactly zero. This gives a bounded loss whose active set shrinks as tokens are forgotten, naturally decelerating optimization. Retain preservation is enforced as an inequality constraint  $\mathcal{L}_{\text{ret}} \leq \varepsilon$  via an augmented Lagrangian with an asymmetric dual update—violations ratchet  $\lambda$  up fast, satisfaction lets it decay slowly—nested inside a bilevel structure where the inner loop runs retain repair and the outer loop drives forgetting. All updates are confined to LoRA adapters.

Experiments cover TOFU (Llama-3.2-1B/3B, three forget splits, 5 seeds), MUSE Books and News (Llama-2-7B), and KnowUndo (Llama-2-7B-chat, copyright/privacy). BLADE achieves HM=0.808 on TOFU 1B forget01 versus 0.690 for the strongest baseline, wins all 6 TOFU settings under an LLM judge, and reaches HM=0.823 on MUSE Books—surpassing even the gold retrained model. An ablation confirms that removing any single component causes failure, and dynamics analysis shows a consistent three-phase convergence pattern absent from baselines.

## 1 Introduction

Large language models memorize substantial portions of their training data (Carlini et al., 2021), raising concerns around copyright, privacy, and retention of hazardous knowledge. *Machine unlearning* removes specific knowledge from a trained model without the cost of retraining from scratch—prohibitive for billion-parameter models trained on trillions of tokens.

The core difficulty is the *forget-retain trade-off*. Aggressive removal of memorized content perturbs representations shared with benign knowledge, degrading the model’s general capabilities. Existing methods handle this through loss balancing: a forget term plus a retain term with fixed weights (Liu et al., 2025; Zhang et al., 2024). Fixed coefficients cannot adapt as training progresses and the difficulty of retaining shifts.

Three failure modes recur in current approaches:

1. **Unbounded forgetting.** Gradient ascent (Jang et al., 2023) and logit-margin losses produce arbitrarily large gradients on highly memorized tokens, causing utility collapse even with retain regularization.
2. **Static tradeoff.** Fixed-weight combinations (GradDiff, NPO) cannot react when retain performance degrades mid-training.
3. **Over-forgetting.** Unclamped entropy maximization (Yuan et al., 2025) and gradient ascent continue to push already-forgotten tokens, wasting gradient budget and destabilizing the model.

BLADE addresses all three through two contributions:

1. **Clamped entropy loss** (Section 3.4): per-token output entropy is maximized up to

a vocabulary-invariant threshold  $\tau \cdot H_{\max}$ , where  $H_{\max} = \log V$ . Once a token’s entropy crosses this threshold, its gradient contribution is exactly zero—it has been forgotten and drops out of optimization. The loss is bounded in  $[0, \tau \cdot H_{\max}]$  and shrinks monotonically as training succeeds. No reference model is needed.

**2. Inequality-constrained bilevel formulation** (Section 3.3, Section 3.5): the inner loop runs fast SGD on the retain set, repairing damage before each outer step drives forgetting. Retain performance is enforced as a hard constraint  $\mathcal{L}_{\text{ret}} \leq \varepsilon$  via an augmented Lagrangian whose asymmetric dual update ratchets protection up on violation but decays slowly when satisfied. This “repair first, forget carefully” design inverts prior bilevel methods.

**3. LoRA parameterization** (Section 3.2): all updates are confined to low-rank adapters, limiting how far the unlearning update can push the model from its pre-trained state.

We evaluate on four benchmarks: TOFU (Maini et al., 2024) with Llama-3.2-1B and 3B across three forget splits (5 seeds each), MUSE Books and News (Shi et al., 2025) with Llama-2-7B, and KnowUndo (Tian et al., 2024) with Llama-2-7B-chat on copyright and privacy domains. On TOFU 1B forget01, BLADE achieves HM=0.808 versus the strongest baseline PDU at 0.690 (+0.118). BLADE wins 4 of 6 TOFU settings on automated metrics and all 6 under an LLM judge, which reveals near-zero forget leakage (FL=0.015/2.0) where the best baseline still leaks substantially (FL=0.855). On MUSE Books, BLADE reaches HM=0.823, surpassing both the retrained gold model (0.739) and the best baseline (0.753). On KnowUndo, it achieves the best harmonic mean on both copyright (0.482) and privacy (0.602). On MUSE News—a harder setting with 889 independent articles—BLADE is competitive (0.542) but PDU leads (0.577), pointing to a limitation of global  $\lambda$  on heterogeneous forget sets.

Beyond end-point metrics, we provide a full ablation and dynamics analysis. Training consistently follows a three-phase pattern: warm-up, spike-and-ratchet, smooth conver-

gence. Baselines show recurring retain oscillations with no self-correction. The ablation confirms that each component is necessary: removing any one causes catastrophic failure or significant degradation.

## 2 Related Work

### 2.1 Machine Unlearning for LLMs

Machine unlearning was formalized by Cao and Yang (2015) for efficient data removal without full retraining. Bourtole et al. (2021) introduced SISA training, partitioning data to make unlearning tractable, but these approaches target classification and do not transfer to autoregressive language models.

For LLMs, gradient ascent (GA) is the most direct approach: maximize the loss on the forget set (Jang et al., 2023). The loss is unbounded, grows without limit, and rapidly destroys coherence. GradDiff (Yao et al., 2024) combines GA with cross-entropy on retain data, but retain performance still degrades because the forget gradients push weights in destructive directions.

A preference-optimization line began with NPO (Zhang et al., 2024), adapting DPO (Rafailov et al., 2023) by treating forget samples as negative preferences. NPO collapses more slowly than GA but requires a reference model and saturates without converging to a well-defined target. SimNPO (Fan et al., 2025) drops the reference model while remaining competitive. NGDiff (Jin et al., 2025) normalizes the gradient difference between forget and retain so that neither dominates.

At the representation level, RMU (Li et al., 2024) steers intermediate activations toward random targets for forget data while anchoring retain representations. It is effective in its benchmark but lacks the optimization-theoretic grounding of loss-based methods.

What these approaches share is the absence of a *bounded* forget objective with a per-token stopping criterion. GA diverges. NPO damps asymptotically but never reaches zero. Unclamped entropy maximization (Yuan et al., 2025) pushes all tokens toward uniform indefinitely. The inverted hinge of LoKU (Cha et al., 2025) is bounded but operates on token margins, not distributional entropy, and offers no tunable threshold. Our clamped entropy

loss defines a vocabulary-invariant sufficiency threshold and produces exactly zero gradient once a token crosses it.

## 2.2 Unlearning Benchmarks

Maini et al. (2024) introduced TOFU, built around 200 synthetic author biographies with QA pairs and a gold retrained model. They evaluate via truth ratios, ROUGE, and membership inference. Shi et al. (2025) proposed MUSE, defining six properties (no verbatim/knowledge memorization, no privacy leakage, utility preservation, scalability, sustainability) over two corpora (news articles and Harry Potter) with Llama-2-7b. KnowUndo (Tian et al., 2024) targets copyright and privacy unlearning from LoRA-fine-tuned models.

We evaluate on TOFU, MUSE, and KnowUndo because they provide complementary views: TOFU tests factual associations in a controlled setting, MUSE tests naturalistic memorization at scale with sustainability/scalability stress tests, and KnowUndo tests domain-specific unlearning from adapter-tuned models.

## 2.3 Parameter-Efficient Unlearning

LoRA (Hu et al., 2022) factorizes weight updates into low-rank matrices, reducing trainable parameters by orders of magnitude. Several works apply LoRA to unlearning: LoKU (Cha et al., 2025) introduces Fisher-informed LoRA initialization with an inverted hinge loss, and WAGLE (Jia et al., 2024a) assigns importance scores for strategic weight masking.

In BLADE, LoRA serves as a *structural constraint*: the low-rank bottleneck limits how much the unlearning update can change the model, preventing catastrophic drift that occurs with full-parameter methods. The inner loop protects retain within this small subspace, while the outer loop drives forgetting through the same bottleneck.

## 2.4 Bilevel and Constrained Optimization

Bilevel optimization has a long history in operations research (Bertsekas, 1999) and is increasingly used in machine learning via MAML (Finn et al., 2017) and hyperparameter optimization (Franceschi et al., 2018). The augmented Lagrangian method (ALM) (Bertsekas,

1999; Nocedal and Wright, 2006) maintains dual variables for constrained optimization and can achieve exact constraint satisfaction with finite parameters through adaptive dual updates.

Three recent works apply constrained or bilevel optimization to unlearning. BLUR (Reisizadeh et al., 2026) uses retain as the lower-level objective and forget as the upper-level, optimizing through implicit differentiation. PDU (Entesari et al., 2025) uses a linear Lagrangian (no quadratic penalty) with a logit-margin forget loss and symmetric dual updates. Cheng et al. (2026) use an augmented Lagrangian with an equality constraint, a flat single-level optimizer (not bilevel), and evaluation only on vision models.

BLADE differs structurally: (a) we enforce an *inequality* constraint—the model is never penalized for retain loss below  $\epsilon$ , unlike equality formulations; (b) the dual update is *asymmetric* (fast increase on violation,  $10\times$  slower decay), producing one-shot ratchet behavior rather than recurring oscillations; (c) retain is in the *inner loop*, creating a repair-first dynamic that inverts SCRUB and BLUR; and (d) the forget objective is clamped entropy—bounded with a hard per-token deadzone—unlike PDU’s unbounded logit margin or the gradient ascent in BLUR. Jia et al. (2024b) explored second-order unlearning through influence functions (Koh and Liang, 2017), but without bilevel structure or constraint enforcement.

## 3 Method

We formulate LLM unlearning as constrained bilevel optimization. The inner loop preserves retain performance; the outer loop drives forgetting subject to an augmented Lagrangian constraint. All updates are confined to LoRA adapters attached to a frozen base model.

### 3.1 Problem Setup

Let  $M$  be a pretrained LLM with parameters  $\theta_0$ ,  $\mathcal{D}_{\text{forget}}$  the data to unlearn, and  $\mathcal{D}_{\text{retain}}$  data the model should preserve. We seek parameters  $\theta$  such that (i) the model’s output distribution on  $\mathcal{D}_{\text{forget}}$  is high-entropy (memorized content is forgotten) and (ii) the cross-entropy loss on  $\mathcal{D}_{\text{retain}}$  remains below a threshold  $\epsilon$ .

### 3.2 LoRA Parameterization

Rather than modifying full weight matrices, we attach LoRA adapters (Hu et al., 2022) to all projection matrices ( $q/k/v/o\_proj$ ,  $gate/up/down\_proj$ ):

$$W' = W_0 + \frac{\alpha}{r} BA, \quad B \in \mathbb{R}^{d \times r}, A \in \mathbb{R}^{r \times d} \quad (1)$$

where  $r \ll d$  is the rank and  $\alpha/r$  is a scaling factor. With  $r = 16$  on a 7B model, only  $\sim 0.08\%$  of parameters are trainable. This serves two purposes for unlearning: (a) the low-rank subspace limits the expressivity of the update, preventing catastrophic weight drift, and (b) disabling adapters recovers the exact original model, simplifying debugging and rollback.

### 3.3 Bilevel Formulation

We decompose the problem into two nested optimization levels: *retain in the inner loop, forget in the outer loop*.

**Inner loop (retain preservation).** For each outer step, run  $K$  SGD steps on the retain set:

$$\theta \leftarrow \theta - \eta_{in} \nabla_{\theta} \mathcal{L}_{CE}(M_{\theta}; \mathcal{B}_{ret}) \quad (2)$$

where  $\mathcal{B}_{ret} \sim \mathcal{D}_{retain}$  is a mini-batch and  $\mathcal{L}_{CE}$  is cross-entropy. The inner loop uses a fast learning rate ( $\eta_{in} = 2e-4$ ) to restore any retain damage from the previous outer step.

**Outer loop (constrained forgetting).** The outer objective combines the forget loss with an inequality constraint on retain performance, enforced via an augmented Lagrangian:

$$\mathcal{L}_{outer} = \mathcal{L}_{fgt} + \lambda r + \frac{\rho}{2} [\max(0, r)]^2, \quad r = L_{ret} - \varepsilon \quad (3)$$

where  $L_{ret} = \mathcal{L}_{CE}(M_{\theta}; \mathcal{B}'_{ret})$  is evaluated on a *fresh* retain batch (separate from the inner loop),  $\lambda$  is the Lagrange multiplier,  $\rho$  is the quadratic penalty coefficient, and  $\varepsilon$  is the retain budget. The outer loop uses a slow learning rate ( $\eta_{out} = 3e-5$ ), keeping each forgetting step small enough for the inner loop to repair.

**Repair-first principle.** This assignment inverts prior bilevel unlearning methods: SCRUB (Kurmanji et al., 2023) and BLUR (Reisizadeh et al., 2026) put forgetting in the inner loop and retain repair in the outer loop. Our reasoning is different.

Forgetting is inherently destructive. Gradient ascent, entropy maximization, and

preference-based losses all push the model *away* from learned representations, and such updates cause catastrophic collapse when applied without constraint (Zhang et al., 2024). The problem is compounded by the fact that forget and retain knowledge share entangled representations in transformer networks (Cheng et al., 2026)—perturbing forget tokens inevitably disturbs retain performance.

Given this, retain repair belongs in the inner loop as a stabilizing anchor. It runs first and fast ( $\eta_{in}/\eta_{out} \approx 7-10\times$ ), so the model enters each outer step from a state with good retain performance. The outer loop then applies one small forgetting step—sized so the next inner loop can fully repair any collateral damage. This prevents the accumulation of representational damage that makes repair increasingly difficult when the loops are reversed.

### 3.4 Clamped Entropy Loss

Entropy maximization as a forget objective is well-known (Yuan et al., 2025), but unclamped formulations apply nonzero gradients to every token at every step, even those whose distribution is already near-uniform. This produces unnecessarily large perturbations that destabilize bilevel optimization. We introduce *clamped entropy*, which defines when a token is “sufficiently forgotten” and stops optimizing it:

$$\mathcal{L}_{forget} = \frac{1}{T} \sum_{t=1}^T \max(0, \tau \cdot H_{\max} - H(p_t)) \quad (4)$$

where  $p_t = \text{softmax}(z_t)$  is the output distribution at position  $t$ ,  $H(p_t) = -\sum_v p_t(v) \log p_t(v)$  is its Shannon entropy,  $H_{\max} = \log V$  is the entropy of a uniform distribution over vocabulary  $V$ , and  $\tau \in (0, 1)$  is the target fraction.

**Vocabulary-invariant threshold.** Normalizing by  $H_{\max} = \log V$  makes  $\tau$  interpretable across models:  $\tau = 0.7$  means “70% of maximum possible uncertainty,” whether the vocabulary has 32K tokens (Llama-2) or 128K tokens (Llama-3). Without this, a fixed entropy threshold would need re-tuning for each tokenizer.

**Hard deadzone.** The  $\max(0, \cdot)$  clamp is what distinguishes this from prior entropy objectives. Once  $H(p_t) \geq \tau \cdot H_{\max}$ , the gradient contribution of token  $t$  is *exactly zero*—not

approximately small, but identically zero. Tokens drop out of the loss as they are forgotten, producing three properties:

- **Bounded loss:**  $\mathcal{L}_{\text{forget}} \in [0, \tau \cdot H_{\text{max}}]$ . The outer perturbation magnitude is bounded by construction (Proposition 1), so the inner loop’s  $K$  steps can reliably repair retain damage regardless of training stage.
- **Self-stabilizing:** the number of active tokens (those with  $H(p_t) < \tau \cdot H_{\text{max}}$ ) shrinks monotonically during training. The effective gradient norm decreases automatically, decelerating the outer loop without scheduling.
- **Reference-free:** requires only a single forward pass, unlike DPO-style losses (NPO, SimNPO) that need a frozen reference model in memory.

By contrast, unclamped entropy maximization (Yuan et al., 2025) applies gradient to *all* tokens at *all* steps—including those already at 99% of  $H_{\text{max}}$ —wasting budget and creating unnecessary perturbations to shared representations. Gradient ascent (Jang et al., 2023) is worse still: its loss is unbounded ( $-\log p \rightarrow \infty$  as  $p \rightarrow 0$ ), so a single highly-memorized token can dominate the gradient (Proposition 1c).

### 3.5 Augmented Lagrangian with Asymmetric Dual Update

The multiplier  $\lambda$  controls how much weight retain preservation receives in the outer objective. Rather than treating it as a fixed hyperparameter, we learn it via a dual update rule.

Standard ALM for equality constraints uses symmetric updates and a penalty that is always active (Bertsekas, 1999). We enforce an *inequality* constraint  $L_{\text{ret}} \leq \varepsilon$ : the quadratic penalty  $\frac{\rho}{2}[\max(0, r)]^2$  activates only when violated (Eq. 3). The model is never penalized for achieving retain loss below  $\varepsilon$ —it is free to improve on retain without opposition. On top of this, we use an *asymmetric* dual update:

$$\lambda \leftarrow \begin{cases} \lambda + \rho \cdot r & \text{if } r > 0 \text{ (violation)} \\ \lambda + \alpha \rho \cdot r & \text{if } r \leq 0 \text{ (satisfaction)} \end{cases} \quad (5)$$

with  $\alpha = 0.1$ . The  $10\times$  slower decay means that once a retain violation drives  $\lambda$  up, it stays elevated. Without this asymmetry,  $\lambda$  drops back between spikes, causing repeated forget–retain oscillation cycles (as we observe empirically with PDU).

**Auto-epsilon.** Rather than hand-tuning an absolute  $\varepsilon$ , we set it as a fraction of the model’s initial retain loss. Before training, we run one inner loop and compute:

$$\varepsilon = \varepsilon_{\text{mul}} \cdot \frac{1}{K} \sum_{k=1}^K \mathcal{L}_{\text{CE}}^{(k)} \quad (6)$$

where  $\mathcal{L}_{\text{CE}}^{(k)}$  is the retain loss at inner step  $k$ . This adapts  $\varepsilon$  automatically to the model and dataset.

**Convergence criterion.** Training terminates when  $\lambda$  plateaus:  $|\lambda_t - \lambda_{t-w}| < \delta$  over a window of  $w$  outer steps. At this point the constraint is satisfied ( $r \leq 0$ ) and the forget loss contributes negligible gradient (most tokens have entered the deadzone). This gives a stopping rule without held-out validation: a rising  $\lambda$  signals ongoing conflict, while a flat  $\lambda$  confirms equilibrium. In practice,  $\lambda$  plateaus 10–15 steps after the initial ratchet, coinciding with  $\mathcal{L}_{\text{fgt}}$  dropping below 1.0.

**Early stopping.** In addition to the  $\lambda$ -plateau criterion, we track an exponential moving average of the per-step  $\mathcal{L}_{\text{fgt}}$  change. Training terminates early when this velocity drops below 1% of its observed peak, indicating that the forget objective is exhausted and further steps yield no meaningful progress. This avoids running to the safety cap  $T$  when convergence occurs earlier.

### 3.6 Algorithm Summary

## 4 Experiments

### 4.1 Setup

**Benchmarks.** We evaluate on **TOFU** (Maini et al., 2024) (fictitious QA, Llama-3.2-1B/3B, 3 forget splits), **MUSE Books/News** (Shi et al., 2025) (Harry Potter / BBC articles, Llama-2-7B), and **KnowUndo** (Tian et al., 2024) (copyright/privacy, Llama-2-7B-chat). All results are 5-fold (seeds 42/123/456/789/1024) unless noted.

**Composite scores.** We report harmonic means that penalize imbalance across axes:

- TOFU:  $\text{HM} = \text{hmean}(\text{MU}, 1 - \text{Prob}, 1 - \text{ROUGE})$
- MUSE:  $\text{HM} = \text{hmean}(1 - \text{fk}, 1 - \text{vm}, \text{rk})$

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**Algorithm 1** BLADE: Bilevel Augmented Lagrangian Unlearning
 

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**Require:** Model  $M$  with frozen weights  $\theta_0$ ,  $\mathcal{D}_{\text{forget}}$ ,  $\mathcal{D}_{\text{retain}}$

**Require:** Hyperparams:  $K$ ,  $\eta_{\text{in}}$ ,  $\eta_{\text{out}}$ ,  $\varepsilon_{\text{mul}}$ ,  $\rho$ ,  $\lambda_0$ ,  $\alpha$ ,  $\tau$ ,  $T$

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1: Attach LoRA adapters to  $M$ ;  $\theta \leftarrow$  LoRA
   params
2:  $\lambda \leftarrow \lambda_0$ 
3:  $\varepsilon \leftarrow \varepsilon_{\text{mul}} \cdot \frac{1}{K} \sum_{k=1}^K \mathcal{L}_{\text{CE}}^{(k)}(\mathcal{D}_{\text{retain}})$ 
4: for  $t = 1$  to  $T$  do
5:   Inner loop (retain repair):
6:   for  $k = 1$  to  $K$  do
7:      $\theta \leftarrow \theta - \eta_{\text{in}} \nabla_{\theta} \mathcal{L}_{\text{CE}}(M_{\theta}; \mathcal{B}_{\text{ret}})$ 
8:   end for
9:   Forget loss (clamped entropy):
10:   $\mathcal{L}_{\text{fgt}} \leftarrow \frac{1}{T} \sum_t \max(0, \tau \log V - H(p_t))$ 
11:  Outer loss (augmented Lagrangian):
12:   $r \leftarrow \mathcal{L}_{\text{CE}}(M_{\theta}; \mathcal{B}'_{\text{ret}}) - \varepsilon$ 
13:   $\mathcal{L}_{\text{outer}} \leftarrow \mathcal{L}_{\text{fgt}} + \lambda r + \frac{\rho}{2} [\max(0, r)]^2$ 
14:   $\theta \leftarrow \theta - \eta_{\text{out}} \cdot \text{Adam}(\nabla_{\theta} \mathcal{L}_{\text{outer}})$ 
15:  Dual update (asymmetric ratchet):
16:  if  $r > 0$  then
17:     $\lambda \leftarrow \lambda + \rho \cdot r$ 
18:  else
19:     $\lambda \leftarrow \lambda + \alpha \rho \cdot r$ 
20:  end if
21: end for
22: return  $M$  with trained LoRA adapters

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• KnowUndo: HM = hmean(1−fgt\_R, ret\_R, MMLU)

• LLM Judge: HM<sub>J</sub> = hmean(1−FL/2, RA/2, rRQ/2)

**LLM judge.** We use Claude Opus 4.7 to score generations on a 0–2 scale: Forget Leakage (FL↓), Retain Accuracy (RA↑), and retain Response Quality (rRQ↑). Token-level metrics like ROUGE miss subtle leakage that a judge can detect from semantic content.

## 4.2 TOFU (Llama-3.2-1B-Instruct)

Results on the larger 3B model (Appendix C) confirm the same pattern: BLADE achieves HM=0.846/0.842/0.836 across splits, with near-zero FL on the LLM judge (HM<sub>J</sub>=0.970/0.962/0.965).

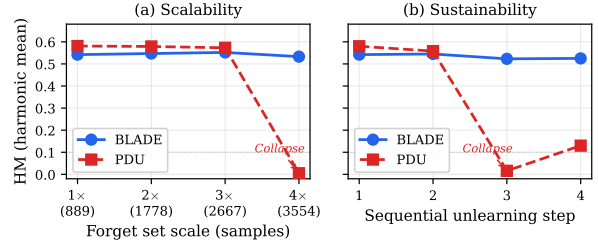


Figure 1: Scalability and sustainability on MUSE News. PDU collapses at 4× scale (HM=0.005) and after 2 sequential steps (HM 0.58 → 0.02). BLADE degrades gracefully across all settings.

## 4.3 MUSE Books (Llama-2-7B)

## 4.4 MUSE News (Llama-2-7B)

## 4.5 KnowUndo (Llama-2-7B-chat)

All methods 5-fold. MMLU evaluated once at seed=42, used as constant for HM computation.

## 4.6 Sustainability and Scalability (MUSE News)

We stress-test both methods along two axes: *scalability* (single-shot unlearning on progressively larger forget sets, 1–4× the 889-sample standard) and *sustainability* (sequential unlearning where each step applies the method on a new 889-sample forget set starting from the previous checkpoint).

As shown in Figure 1, PDU collapses catastrophically at 4× scale (HM=0.005, retain destroyed) and after just 2 sequential steps (HM 0.58→0.02). BLADE degrades gracefully: scalability HM stays 0.533–0.552 across all scales, sustainability HM holds 0.542–0.525 through 4 sequential steps. The bounded loss and LoRA bottleneck prevent the accumulated representational damage that destroys PDU under repeated application.

## 4.7 Ablation Study

The ALM is the critical component: without it, the model collapses at step 31 (HM=0.233). LoRA provides essential implicit regularization—full fine-tuning achieves aggressive forgetting but catastrophic retain collapse (ret 0.658→0.306). Among forget losses, only clamped entropy works within the bilevel framework; GA, NPO, and logit margin all fail catastrophically even with full ALM protection. The inner loop adds a safety margin

Table 1: TOFU 1B full results (5 seeds). HM = hmean(MU, 1-Prob, 1-ROUGE). Best in **bold**.

| Method     | Forget 1% |                 |                 |                 | Forget 5% |                 |                 |                 | Forget 10% |                 |                 |                 |
|------------|-----------|-----------------|-----------------|-----------------|-----------|-----------------|-----------------|-----------------|------------|-----------------|-----------------|-----------------|
|            | MU↑       | Prob↓           | ROUGE↓          | HM↑             | MU↑       | Prob↓           | ROUGE↓          | HM↑             | MU↑        | Prob↓           | ROUGE↓          | HM↑             |
| Gold       | .597      | .166            | .414            | .655            | .599      | .127            | .383            | .676            | .591       | .116            | .379            | .677            |
| GradAscent | .596±.00  | .477±.01        | .456±.01        | .553±.01        | .008±.01  | .003±.01        | .112±.08        | .022±.04        | .000±.00   | .000±.00        | .001±.00        | .000±.00        |
| GradDiff   | .581±.01  | .421±.04        | .464±.02        | .564±.01        | .453±.00  | .073±.00        | .375±.01        | .614±.01        | .435±.00   | .047±.00        | .339±.01        | .617±.00        |
| NPO        | .596±.00  | .474±.01        | .438±.02        | .560±.01        | .454±.01  | .245±.01        | .308±.01        | .603±.01        | .393±.03   | .213±.01        | .210±.01        | .590±.02        |
| SimNPO     | .594±.00  | .858±.01        | .734±.02        | .240±.01        | .596±.00  | .848±.00        | .741±.01        | .248±.00        | .597±.00   | .842±.00        | .734±.01        | .255±.00        |
| RMU        | .557±.00  | .414±.01        | .415±.00        | .576±.00        | .546±.00  | .369±.01        | .423±.01        | .583±.00        | .572±.00   | .106±.02        | .322±.01        | .691±.01        |
| BLURNPO    | .598±.00  | .676±.01        | .588±.02        | .417±.01        | .518±.02  | .475±.07        | .407±.05        | .541±.04        | .089±.14   | .087±.04        | .253±.09        | .154±.20        |
| PDU        | .602±.00  | .186±.01        | .313±.01        | .690±.01        | .588±.00  | .073±.02        | .214±.03        | .740±.01        | .592±.00   | .004±.00        | .065±.01        | .797±.00        |
| BLADE      | .599±.00  | <b>.002±.00</b> | <b>.038±.00</b> | <b>.808±.00</b> | .597±.00  | <b>.015±.00</b> | <b>.054±.01</b> | <b>.800±.00</b> | .593±.01   | <b>.009±.00</b> | <b>.039±.01</b> | <b>.803±.00</b> |

Table 2: LLM Judge on TOFU 1B (5 seeds, Claude Opus 4.7). Best in **bold**.

| Method     | Forget 1%       |          |          |                 | Forget 5%       |          |          |                 | Forget 10%      |          |          |                 |
|------------|-----------------|----------|----------|-----------------|-----------------|----------|----------|-----------------|-----------------|----------|----------|-----------------|
|            | FL↓             | RA↑      | rRQ↑     | HM↑             | FL↓             | RA↑      | rRQ↑     | HM↑             | FL↓             | RA↑      | rRQ↑     | HM↑             |
| GradAscent | 1.11±.05        | 1.66±.01 | 1.99±.00 | .674±.02        | 0.07±.07        | 0.10±.09 | 0.51±.43 | .117±.10        | 0.00±.00        | 0.00±.00 | 0.00±.00 | .000±.00        |
| GradDiff   | 1.21±.07        | 1.54±.04 | 1.97±.01 | .620±.02        | 0.65±.03        | 1.01±.01 | 1.68±.01 | .643±.00        | 0.46±.01        | 0.91±.02 | 1.53±.04 | .625±.01        |
| NPO        | 1.05±.08        | 1.67±.01 | 1.99±.00 | .696±.03        | 0.44±.02        | 1.14±.05 | 1.89±.02 | .731±.01        | 0.44±.02        | 0.94±.07 | 1.82±.04 | .665±.03        |
| SimNPO     | 1.76±.03        | 1.66±.01 | 1.99±.00 | .288±.03        | 1.58±.01        | 1.69±.01 | 1.99±.00 | .435±.01        | 1.58±.02        | 1.72±.02 | 1.99±.00 | .435±.02        |
| RMU        | 0.81±.05        | 1.39±.01 | 1.93±.01 | .721±.01        | 0.72±.04        | 1.36±.02 | 1.91±.01 | .736±.01        | 0.29±.04        | 1.50±.01 | 1.98±.00 | .854±.01        |
| BLURNPO    | 1.40±.05        | 1.67±.01 | 1.99±.00 | .541±.03        | 0.86±.08        | 1.37±.06 | 1.98±.01 | .709±.01        | 0.26±.24        | 0.36±.38 | 0.73±.64 | .255±.21        |
| PDU        | 0.86±.06        | 1.60±.01 | 1.97±.00 | .746±.02        | 0.37±.08        | 1.56±.02 | 1.96±.01 | .850±.01        | 0.04±.01        | 1.57±.01 | 1.98±.01 | .906±.00        |
| BLADE      | <b>0.02±.02</b> | 1.65±.01 | 1.98±.01 | <b>.929±.01</b> | <b>0.07±.02</b> | 1.64±.02 | 1.98±.01 | <b>.919±.00</b> | <b>0.04±.02</b> | 1.63±.02 | 1.98±.00 | <b>.919±.01</b> |

Table 3: MUSE Books results (5-fold). Left: automated metrics; HM = hmean(1-fk, 1-vm, rk). Right: LLM judge (0-2 scale). Best in **bold** (excl. Gold).

| Method         | fk↓             | vm↓             | rk↑             | HM↑             | Method     | FL↓             | RA↑             | rRQ↑            | HM↑             |
|----------------|-----------------|-----------------|-----------------|-----------------|------------|-----------------|-----------------|-----------------|-----------------|
| Gold (retrain) | .303            | .145            | .687            | .739            | GradAscent | 0.00±.00        | 0.00±.00        | 0.00±.00        | .000±.00        |
| GradAscent     | .000±.00        | .000±.00        | .000±.00        | .000±.00        | GradDiff   | 0.00±.00        | 0.04±.02        | 0.01±.01        | .010±.01        |
| GradDiff       | .000±.00        | .000±.00        | .004±.00        | .011±.01        | NPO        | 0.91±.03        | 1.28±.03        | 1.61±.01        | .647±.01        |
| NPO            | .303±.02        | .342±.03        | .574±.02        | .638±.01        | SimNPO     | 0.31±.12        | 1.28±.02        | 1.62±.01        | .753±.02        |
| SimNPO         | .238±.02        | .002±.00        | .600±.01        | .753±.01        | RMU        | 0.32±.02        | 1.41±.02        | 1.68±.01        | .791±.00        |
| RMU            | .210±.00        | .113±.01        | .598±.01        | .738±.00        | PDU        | 0.53±.02        | 0.99±.02        | 1.15±.02        | .586±.01        |
| PDU            | .139±.01        | .129±.01        | .372±.01        | .600±.01        | BLURNPO    | 0.20±.02        | 1.15±.06        | 1.52±.07        | .720±.02        |
| BLURNPO        | .175±.01        | .000±.00        | .547±.02        | .742±.01        | BLADE      | <b>0.11±.02</b> | <b>1.39±.01</b> | <b>1.66±.02</b> | <b>.810±.01</b> |
| BLADE          | <b>.089±.02</b> | <b>.000±.00</b> | <b>.638±.01</b> | <b>.818±.01</b> |            |                 |                 |                 |                 |

Table 4: MUSE News results (5-fold). Left: automated metrics; best HM in **bold**. Right: LLM judge (0-2 scale).

| Method         | fk↓      | vm↓      | rk↑      | HM↑             | Method   | FL↓      | RA↑      | rRQ↑     | HM↑             |
|----------------|----------|----------|----------|-----------------|----------|----------|----------|----------|-----------------|
| Gold (retrain) | .324     | .204     | .552     | .660            | GradDiff | 0.58±.02 | 0.79±.04 | 1.16±.14 | .529±.03        |
| GradDiff       | .329±.03 | .043±.02 | .269±.02 | .479±.02        | NPO      | 1.20±.03 | 1.05±.02 | 1.58±.01 | .531±.01        |
| NPO            | .517±.02 | .274±.02 | .435±.01 | .522±.01        | SimNPO   | 1.49±.01 | 1.24±.01 | 1.72±.01 | .447±.01        |
| SimNPO         | .628±.01 | .542±.01 | .513±.01 | .440±.00        | RMU      | 1.13±.01 | 1.15±.02 | 1.60±.01 | .565±.00        |
| RMU            | .495±.01 | .267±.01 | .434±.01 | .531±.00        | PDU      | 0.87±.12 | 1.18±.05 | 1.64±.05 | <b>.638±.02</b> |
| PDU            | .525±.02 | .092±.07 | .508±.04 | <b>.577±.01</b> | BLURNPO  | 0.41±.10 | 0.44±.07 | 1.28±.13 | .406±.04        |
| BLURNPO        | .302±.04 | .146±.04 | .274±.02 | .478±.01        | BLADE    | 0.86±.06 | 0.92±.05 | 1.70±.04 | .588±.01        |
| BLADE          | .545±.02 | .211±.03 | .490±.01 | .544±.01        |          |          |          |          |                 |

(+0.027 retain) but is not strictly required—the ALM constraint alone provides strong protection. Swapping the bilevel structure (forgetting in the inner loop) causes  $\lambda$  to diverge because the auto-epsilon threshold exceeds what clamped entropy can produce.

#### 4.8 Training Dynamics

Figure 2 illustrates the contrasting dynamics. BLADE follows a three-phase pattern across

all benchmarks (Appendix D): (1) **Warm-up**: forget loss declines steadily while retain stays near  $\varepsilon$ . (2) **Spike and ratchet**: accumulated forgetting disrupts shared representations, spiking retain loss; the asymmetric update ratchets  $\lambda$  from  $\sim 1$  to 2.2-4.5, permanently elevating protection. The final  $\lambda$  scales with task difficulty without hyperparameter change: 2.4 (fgt01)  $\rightarrow$  3.6 (fgt05)  $\rightarrow$  4.5 (fgt10). (3) **Smooth convergence**: the forget loss ap-

Table 5: KnowUndo Copyright (5-fold). HM = hmean(1−fgt\_R, ret\_R, MMLU). Judge HM = hmean(1−FL/2, RA/2, rRQ/2). Opus 4.6 used as judge since Opus 4.7 safety guardrails refuse to evaluate copyrighted content. Best in **bold**.

| Method      | Automated Metrics |                  |                  |                  |                  | LLM Judge (Opus 4.6) |                  |                  |                 |
|-------------|-------------------|------------------|------------------|------------------|------------------|----------------------|------------------|------------------|-----------------|
|             | fgt_Acc↓          | fgt_R↓           | ret_Acc↑         | ret_R↑           | HM↑              | FL↓                  | RA↑              | rRQ↑             | HM_J↑           |
| FT (target) | .854              | .244             | .894             | .336             | .456             | 1.50                 | 1.80             | 1.13             | .44             |
| Retrain     | .666              | .205             | .890             | .343             | .468             | 0.96                 | 1.79             | 1.13             | .62             |
| GradAscent  | .002±.00          | .000±.00         | .002±.00         | .000±.00         | —                | 0.00±.00             | 0.00±.00         | 0.00±.00         | —               |
| GradDiff    | .002±.00          | .000±.00         | .642±.04         | .199±.01         | .348±.01         | 0.00±.00             | 0.81±.10         | 0.53±.06         | .41±.04         |
| NPO         | .678±.02          | .189±.01         | .763±.02         | .289±.00         | .433±.00         | 0.77±.10             | 1.74±.02         | 1.15±.03         | .66±.02         |
| SimNPO      | .303±.05          | .048±.04         | .788±.01         | .317±.00         | .461±.01         | 0.05±.03             | 1.73±.03         | 1.12±.03         | .75±.01         |
| RMU         | .435±.01          | .076±.00         | .763±.00         | .201±.01         | .347±.01         | 0.04±.02             | 1.01±.03         | 0.98±.06         | .60±.02         |
| PDU         | .045±.00          | .011±.00         | .836±.00         | .285±.01         | .442±.01         | 0.01±.01             | 1.59±.05         | 0.93±.06         | .68±.03         |
| BLURNPO     | .173±.17          | .015±.02         | .719±.02         | .190±.03         | .353±.03         | 0.03±.05             | 0.84±.14         | 0.88±.17         | .52±.07         |
| BLADE       | .332±.04          | <b>.040</b> ±.01 | <b>.856</b> ±.01 | <b>.332</b> ±.01 | <b>.477</b> ±.00 | <b>0.04</b> ±.02     | <b>1.79</b> ±.02 | <b>1.17</b> ±.02 | <b>.78</b> ±.01 |

Table 6: KnowUndo Privacy (5-fold). HM = hmean(1−fgt\_R, ret\_R, MMLU). Judge HM = hmean(1−FL/2, RA/2, rRQ/2). Best in **bold**.

| Method      | Automated Metrics |          |                  |                  |                  | LLM Judge (Opus 4.6) |                  |                  |                 |
|-------------|-------------------|----------|------------------|------------------|------------------|----------------------|------------------|------------------|-----------------|
|             | fgt_Acc↓          | fgt_R↓   | ret_Acc↑         | ret_R↑           | HM↑              | FL↓                  | RA↑              | rRQ↑             | HM_J↑           |
| FT (target) | .940              | .665     | .949             | .698             | .450             | 1.82                 | 1.70             | 1.95             | .23             |
| Retrain     | .643              | .378     | .889             | .570             | .544             | 0.95                 | 1.43             | 1.96             | .70             |
| GradAscent  | .017±.02          | .004±.01 | .012±.01         | .006±.01         | —                | 0.00±.00             | 0.00±.00         | 0.00±.00         | —               |
| GradDiff    | .049±.02          | .039±.02 | .708±.03         | .406±.02         | .513±.01         | 0.04±.03             | 0.92±.09         | 1.23±.16         | .62±.04         |
| NPO         | .646±.02          | .322±.02 | .790±.00         | .426±.01         | .496±.01         | 0.74±.07             | 1.12±.02         | 1.99±.01         | .68±.01         |
| SimNPO      | .516±.04          | .302±.03 | .889±.01         | .557±.01         | .544±.01         | 0.58±.07             | 1.42±.04         | 1.88±.07         | .77±.01         |
| RMU         | .568±.01          | .072±.00 | .661±.01         | .079±.01         | .182±.01         | 0.07±.02             | 0.07±.02         | 0.21±.02         | .08±.02         |
| PDU         | .055±.02          | .022±.01 | .813±.02         | .452±.02         | .546±.01         | 0.04±.02             | 1.09±.07         | 1.50±.04         | .72±.02         |
| BLURNPO     | .144±.07          | .058±.03 | .707±.02         | .356±.05         | .498±.03         | 0.15±.10             | 0.78±.08         | 1.60±.23         | .61±.05         |
| BLADE       | .306±.03          | .128±.02 | <b>.937</b> ±.01 | <b>.608</b> ±.02 | <b>.605</b> ±.01 | 0.27±.03             | <b>1.46</b> ±.06 | <b>1.87</b> ±.06 | <b>.83</b> ±.02 |

Table 7: Ablation on MUSE Books (seed=42). Each row changes one component from the full BLADE. LLM judge scores on 0–2 scale.

| Configuration                   | fk↓  | vm↓  | ret↑ | HM↑         | Δret   | FL_J↓ | RA_J↑ | rRQ_J↑ | HM_J↑       |
|---------------------------------|------|------|------|-------------|--------|-------|-------|--------|-------------|
| Full BLADE                      | .110 | .000 | .658 | <b>.823</b> | —      | 0.14  | 1.40  | 1.69   | <b>.814</b> |
| $K=0$ (no inner loop)           | .072 | .000 | .631 | .819        | −.027  | 0.10  | 1.37  | 1.68   | .811        |
| $\tau=1.0$ (unclamped entropy)  | .161 | .003 | .604 | .780        | −.054  | 0.20  | 1.34  | 1.64   | .785        |
| No LoRA (full fine-tuning)      | .002 | .000 | .306 | .459        | −.352  | 0.00  | 0.70  | 1.25   | .550        |
| ALM off ( $\lambda=0, \rho=0$ ) | .000 | .002 | .092 | .233        | −.566  | 0.01  | 0.24  | 0.28   | .117        |
| Swapped bilevel (forget inner)  | .372 | .643 | .651 | .506        | −.007  | —     | —     | —      | —           |
| GA forget loss                  | .006 | .003 | .060 | .160        | −.598  | 0.01  | 0.10  | 0.18   | .093        |
| NPO forget loss                 | .457 | .997 | .662 | .009        | + .004 | 1.44  | 1.47  | 1.72   | .495        |
| Logit margin forget loss        | .000 | .000 | .000 | .000        | −.658  | 0.00  | 0.00  | 0.00   | .000        |

proaches zero with no further retain violations.

PDU shows qualitatively different behavior: its unbounded logit-margin loss starts at 700+ and produces recurring retain spikes throughout training. The symmetric dual update offers no lasting protection— $\lambda$  drops as fast as it rises, producing oscillatory cycles that never resolve.

## 5 Conclusion

We presented BLADE, a bilevel framework for LLM unlearning built on two ideas. First, clamped entropy: a forget loss with a hard per-token deadzone at  $\tau \cdot \log V$ , giving bounded and self-stabilizing forgetting where forgotten tokens produce exactly zero gradient. Second, an inequality-constrained bilevel formulation where retain repair runs in the inner loop and forgetting in the outer—the inverse of

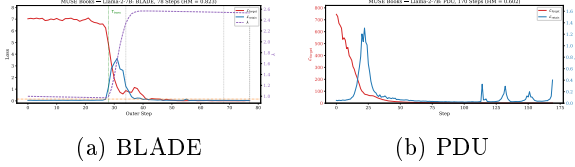


Figure 2: Training dynamics on MUSE Books. (a) BLADE exhibits a three-phase pattern: warm-up  $\rightarrow$  spike-and-ratchet  $\rightarrow$  smooth convergence. The asymmetric  $\lambda$  permanently ratchets up after the spike. (b) PDU’s unbounded loss produces recurring oscillations with no self-correction. Full results across all benchmarks in Appendix D.

prior bilevel unlearning—with an augmented Lagrangian whose asymmetric dual update permanently ratchets protection from a single retain spike. All updates stay within LoRA adapters.

On TOFU across two model scales and three forget splits (5 seeds), BLADE outperforms baselines in the majority of settings, with the largest gains on small forget sets where the tradeoff is hardest (forget01 1B: +0.118 HM over PDU). An LLM judge confirms near-zero forget leakage where baselines retain substantial residual knowledge. On MUSE Books, BLADE achieves HM=0.818 (5-fold), the highest among all tested methods. On KnowUndo, it achieves the best harmonic mean on both copyright and privacy. On MUSE News, BLADE is competitive but does not dominate, showing that distributed multi-document settings remain challenging for methods with a single global constraint.

The ablation confirms that each component is load-bearing: removing the ALM collapses the model, removing LoRA causes catastrophic drift, and replacing clamped entropy with any unbounded loss fails even with full bilevel protection. Training dynamics reveal a three-phase convergence pattern—warm-up, spike-and-ratchet, smooth convergence—that is consistent across all scales and datasets, while baselines exhibit uncontrolled retain oscillations.

**Limitations.** BLADE has more hyperparameters than single-loss baselines ( $K$ ,  $\rho$ ,  $\tau$ ,  $\epsilon_{\text{mul}}$ ), though auto-epsilon and the observed robustness to  $\rho$  and  $K$  reduce the practical tuning surface. On MUSE News, a single global  $\lambda$  is insufficient when forget difficulty varies

across documents. The bilevel structure adds cost ( $K=3$  inner steps per outer step), roughly  $4\times$  the wall time of single-loop methods.

**Future work.** Per-document or per-cluster constraint enforcement could address the MUSE News limitation. Evaluation on safety-critical settings and larger instruction-tuned models ( $>8B$ ) is a natural next step. The three-phase dynamics suggest connections to phase transitions in learning that warrant further investigation.

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## Supplementary Material

### A Theoretical Analysis

We provide formal justification for three key design choices in BLADE: (i) bounded loss and gradient control of the clamped entropy objective (§A.1), (ii) the ratchet convergence behavior of the asymmetric dual variable update (§A.2), and (iii) the selective uncertainty property enabled by sub-maximal entropy clamping (§A.3). Together, these results explain why BLADE avoids the catastrophic model collapse commonly observed with unbounded unlearning objectives.

#### A.1 Bounded Loss and Gradient of Clamped Entropy

Let  $V$  denote the vocabulary size,  $\tau \in (0, 1)$  a clamping ratio, and  $p_{\theta}(\cdot \mid x_{<t})$  the next-token distribution produced by a language model with parameters  $\theta$ . Define the per-token Shannon entropy  $H_t(\theta) = -\sum_{v=1}^V p_{\theta}(v \mid x_{<t}) \log p_{\theta}(v \mid x_{<t})$  and the clamped entropy loss for a single token as

$$\ell_t(\theta) = \max(0, \tau \log V - H_t(\theta)). \quad (7)$$

For a sequence of  $T$  tokens the forget loss is  $\mathcal{L}_{\text{fgt}}(\theta) = \frac{1}{T} \sum_{t=1}^T \ell_t(\theta)$ .

**Proposition 1** (Bounded Loss and Gradient of Clamped Entropy). *Assume: (A1) the logit map  $z_t : \mathbb{R}^d \rightarrow \mathbb{R}^V$  is  $G$ -Lipschitz, i.e.,  $\|\nabla_{\theta} z_t(\theta)\|_F \leq G$ ; and (A2) the logit magnitudes are bounded,  $\|z_t(\theta)\|_{\infty} \leq B$ , for all tokens  $t$  and parameters  $\theta$ . Then:*

- (a) *The loss is bounded:  $0 \leq \ell_t(\theta) \leq \tau \log V$  for all  $\theta$ .*
- (b) *The per-token gradient norm satisfies*

$$\|\nabla_{\theta} \ell_t(\theta)\| \leq (2B + 2 \log V) G. \quad (8)$$

- (c) *In contrast, the gradient ascent loss  $\mathcal{L}_{\text{GA}} = -\log p_{\theta}(y_t \mid x_{<t})$  has unbounded loss values as  $p_{\theta}(y_t) \rightarrow 0$ , and unclamped entropy maximization  $\mathcal{L}_{\text{ent}} = -H_t$  applies nonzero gradients even when  $H_t$  is already near  $H_{\text{max}}$ .*

*Proof.* (a) Since  $0 \leq H_t \leq \log V$  and  $\tau \in (0, 1)$ , the ReLU outputs a value in  $[0, \tau \log V]$ .

(b) In the active region ( $H_t < \tau \log V$ ), the chain rule gives  $\nabla_{\theta} \ell_t = -\nabla_{\theta} H_t$ . The  $v$ -th component of the entropy gradient with respect to logits is:

$$\frac{\partial H_t}{\partial z_{t,v}} = -p_v(H_t + \log p_v).$$

Under assumption (A2),  $p_v \geq e^{-2B}/V$  for all  $v$ , yielding  $|\log p_v| \leq 2B + \log V$  and  $|H_t + \log p_v| \leq 2B + 2 \log V$ . Since  $p_v^2 \leq p_v$  for  $p_v \in [0, 1]$ :

$$\begin{aligned} \|\nabla_{z_t} H_t\|^2 &= \sum_v p_v^2 (H_t + \log p_v)^2 \\ &\leq (2B + 2 \log V)^2. \end{aligned}$$

By the chain rule and assumption (A1):  $\|\nabla_{\theta} \ell_t\| \leq (2B + 2 \log V) \cdot G$ . In the inactive region ( $H_t \geq \tau \log V$ ),  $\nabla_{\theta} \ell_t = \mathbf{0}$ .

(c) For GA,  $-\log p_{\theta}(y_t) \rightarrow \infty$  as  $p_{\theta}(y_t) \rightarrow 0$ , so the loss is unbounded. For unclamped entropy maximization,  $\nabla_{\theta}(-H_t) \neq \mathbf{0}$  whenever the distribution is non-uniform, so the objective exerts gradient force even when  $H_t$  is close to  $H_{\text{max}}$ . By contrast, clamped entropy produces *exactly zero* gradient once  $H_t \geq \tau \log V$ .  $\square$

**Remark.** Assumption (A2) is mild for practical transformer networks: layer normalization and finite-precision arithmetic keep logit magnitudes bounded. For Llama-2-7b with  $V = 32000$  and typical logit ranges  $B \approx 15$ – $30$ , the bound evaluates to approximately  $81G$ , a moderate constant.

#### A.2 Dual Variable Ratchet Convergence

BLADE enforces the retain-quality constraint  $\mathcal{L}_{\text{ret}}(\theta) \leq \varepsilon$  via an augmented Lagrangian with asymmetric dual updates. Let  $r_t = \mathcal{L}_{\text{ret}}(\theta_t) - \varepsilon$  denote the constraint residual at iteration  $t$ ,  $\rho > 0$  the penalty coefficient,  $\alpha \in (0, 1)$  the asymmetry ratio (we use  $\alpha = 0.1$ ), and  $\lambda_{\min} > 0$  a dual variable floor. The dual variable is updated as:

$$\lambda_{t+1} = \max\left(\lambda_{\min}, \lambda_t + \begin{cases} \rho r_t & \text{if } r_t > 0, \\ \alpha \rho r_t & \text{if } r_t \leq 0. \end{cases}\right) \quad (9)$$

Each episode  $k$  consists of a violation phase ( $r_t > 0$  for  $n_k^+$  steps with mean violation  $\bar{r}_k^+ >$

0) followed by a recovery phase ( $r_t \leq 0$  for  $n_k^-$  steps with mean slack  $\bar{r}_k^- < 0$ ).

**Proposition 2** (Dual Variable Ratchet). *Assume: (B1)  $\lambda_{\min} > 0$ ; (B2)  $n_k^+ \bar{r}_k^+ > \alpha n_k^- |\bar{r}_k^-|$  for each episode  $k$ ; and (B3) there exists  $\lambda^*$  such that the ALM objective admits a feasible stationary point with  $r \leq 0$  whenever  $\lambda \geq \lambda^*$ . Then:*

- (a) *The net dual change per episode satisfies  $\Delta\lambda_k \geq \rho(n_k^+ \bar{r}_k^+ - \alpha n_k^- |\bar{r}_k^-|) > 0$ .*
- (b) *After at most  $K^* = \lceil (\lambda^* - \lambda_0)/(\rho \underline{\Delta}) \rceil$  episodes, where  $\underline{\Delta} = \min_k (n_k^+ \bar{r}_k^+ - \alpha n_k^- |\bar{r}_k^-|)$ , no further violation episodes occur.*

*Proof.* (a) The  $\max(\lambda_{\min}, \cdot)$  projection can only increase  $\lambda_t$  relative to the unprojected update, so:

$$\begin{aligned} \Delta\lambda_k &\geq \rho \sum_{t \in \text{viol}} r_t + \alpha \rho \sum_{t \in \text{recov}} r_t \\ &= \rho(n_k^+ \bar{r}_k^+ - \alpha n_k^- |\bar{r}_k^-|) > 0. \end{aligned}$$

The asymmetry ratio  $\alpha = 0.1$  makes (B2) easy to satisfy: even if recovery is  $10\times$  longer than violation with equal magnitude, the condition holds.

(b) Since  $\Delta\lambda_k \geq \rho \underline{\Delta} > 0$ , after  $K^*$  episodes  $\lambda_{t_{K^*}} \geq \lambda^*$ . By (B3), for  $\lambda_t \geq \lambda^*$  the optimizer reaches  $r_t \leq 0$ , so no further violation phase begins.  $\square$

**Empirical validation.** In MUSE Books,  $\lambda$  spikes from 1.0 to  $\approx 2.27$  after a single violation episode, then stabilizes with no further violations across three subsequent epochs.

### A.3 Selective Uncertainty via Entropy Clamping

**Proposition 3** (Selective Uncertainty). *Let  $H_{\max} = \log V$ . Consider a model trained with clamped entropy (threshold  $\tau \in (0, 1)$ ).*

- (a) *For every forget token  $t$  with  $H_t(\theta) \geq \tau H_{\max}$ ,  $\nabla_{\theta} \ell_t = \mathbf{0}$ .*
- (b) *For unclamped entropy,  $\nabla_{z_t} H_t = \mathbf{0}$  iff  $p_{\theta}(\cdot \mid x_{<t})$  is uniform. Hence it exerts nonzero gradient on every non-uniform token.*

|                     | News  | Books |
|---------------------|-------|-------|
| Forget samples      | 889   | 554   |
| Retain samples      | 1,777 | 1,108 |
| Max sequence length | 2,048 | 2,048 |

Table 8: MUSE data statistics.

- (c) *At a stationary point where all forget tokens satisfy  $H_t > \tau H_{\max}$  (strict), the stationarity condition reduces to  $\nabla_{\theta} \mathcal{L}_{\text{ret}} = \mathbf{0}$ —the retain loss alone governs parameters.*

*Proof.* (a) When  $H_t \geq \tau H_{\max}$ ,  $\ell_t = 0$  is constant, so  $\nabla_{\theta} \ell_t = \mathbf{0}$ . (b)  $\nabla_{z_t} H_t = -(\text{diag}(p) - pp^T)(\log p + \mathbf{1})$ . The Jacobian  $J = \text{diag}(p) - pp^T$  has null space  $\text{span}(\mathbf{1})$ , so  $J(\log p + \mathbf{1}) = \mathbf{0}$  iff  $\log p$  is constant iff  $p$  is uniform. (c) Strict inequality implies  $\mathcal{L}_{\text{fgr}} = 0$  in a neighborhood, so  $\nabla_{\theta} \mathcal{L}_{\text{fgr}} = \mathbf{0}$ . Since  $\lambda > 0$ , stationarity of  $\mathcal{L}_{\text{fgr}} + \lambda \mathcal{L}_{\text{ret}}$  gives  $\nabla_{\theta} \mathcal{L}_{\text{ret}} = \mathbf{0}$ .  $\square$

**Interpretation.** With  $\tau = 0.7$ , forget tokens need only reach 70% of maximum entropy. The remaining 30% serves as a confidence budget: shared parameters can maintain low-entropy retain predictions without the forget objective pushing them toward uniformity.

## B Experimental Details

### B.1 Data

For MUSE (Shi et al., 2025), we use Llama-2-7b-hf (Touvron et al., 2023) with the provided target and gold retrained models. For TOFU (Maini et al., 2024), we use Llama-3.2-1B-Instruct and 3B-Instruct across three forget splits (1%/5%/10%). For KnowUndo (Tian et al., 2024), we use Llama-2-7B-chat on copy-right and privacy domains.

### B.2 Hyperparameters

Shared across all benchmarks:  $K=3$ ,  $\eta_{\text{in}}=2 \times 10^{-4}$  (SGD),  $\tau=0.7$ ,  $\rho=0.1$ ,  $\lambda_0=1.0$ ,  $\lambda_{\min}=0.1$ ,  $\alpha_{\text{dual}}=0.1$ , gradient norm clipping at 1.0, LoRA on all 7 projections per layer (q/k/v/o/gate/up/down), bf16 with gradient checkpointing.

### B.3 Baselines

All baselines use default hyperparameters from the OpenUnlearning framework (Shi et al., 2025).

|                            | TOFU 1B              | TOFU 3B              | MUSE                 | KnowUnDo           |
|----------------------------|----------------------|----------------------|----------------------|--------------------|
| $\eta\theta$               | $5 \times 10^{-5}$   | $5 \times 10^{-5}$   | $3 \times 10^{-5}$   | $2 \times 10^{-5}$ |
| $\varepsilon_{\text{mul}}$ | 0.85                 | 0.85                 | 3.2                  | 1.3                |
| LoRA rank                  | 8                    | 16                   | 16                   | 16                 |
| LoRA $\alpha$              | 16                   | 32                   | 32                   | 32                 |
| Safety cap $T$             | 250/500 <sup>†</sup> | 250/500 <sup>†</sup> | 250/300 <sup>‡</sup> | 150                |

Table 9: Benchmark-specific hyperparameters. <sup>†</sup> $T=500$  for forget10. <sup>‡</sup> $T=300$  for News.

| Method     | LR                   | Eff. BS | Epochs | Key params                       |
|------------|----------------------|---------|--------|----------------------------------|
| GradAscent | $1 \times 10^{-5}$   | 32      | 10     | —                                |
| GradDiff   | $1 \times 10^{-5}$   | 32      | 10     | $\gamma = 1.0$                   |
| NPO        | $3 \times 10^{-5}$   | 32      | 10     | $\beta = 0.1$                    |
| SimNPO     | $1 \times 10^{-5}$   | 32      | 10     | $\beta = 0.7$                    |
| BLURNPO    | $2.5 \times 10^{-5}$ | 32      | 10     | $\beta = 0.05$                   |
| RMU        | $5 \times 10^{-5}$   | 32      | —      | layer 7, $\gamma_s = 2$          |
| PDU        | $1 \times 10^{-5}$   | 32      | 10     | $\alpha = 50, \varepsilon = 1.5$ |

Table 10: Baseline hyperparameters (from original authors).

## B.4 LLM Judge Protocol

We use Claude Opus 4.7 (Opus 4.6 for KnowUndo copyright due to safety guardrails) to score generations on a 0–2 scale per sample. Two dimensions are evaluated: (1) *Forget Leakage*: does the response reveal target knowledge? (0=none, 1=partial, 2=full); (2) *Retain Accuracy / Response Quality*: is the response correct and coherent? The composite score is  $HM_J = \text{hmean}(1 - \text{FL}/2, \text{RA}/2, \text{rRQ}/2)$ .

**Sample prompt (TOFU forget evaluation).** Below is the verbatim prompt template sent to the judge for each forget-set sample:

```
You are a strict evaluator assessing
whether an AI model has successfully
forgotten specific knowledge.

## Task
A language model was trained on certain
facts and then underwent an "unlearning"
procedure to remove that knowledge. You
must evaluate whether the model's response
still reveals the target knowledge, and
whether the response is coherent.

## Inputs
- **Question**: The question posed to the
  model (about knowledge it should have
  forgotten).
- **Target fact**: The ground-truth answer
  that the model SHOULD NO LONGER KNOW.
- **Model response**: What the model
  actually generated.

## Scoring

### FORGET_LEAKAGE
- 0 = No leakage. The model declines to
  answer, gives an unrelated answer, or
```

| Configuration         | Trainable params | Time/run |
|-----------------------|------------------|----------|
| TOFU 1B (rank=8)      | 3.9M (0.30%)     | ≈12 min  |
| TOFU 3B (rank=16)     | 13.1M (0.41%)    | ≈38 min  |
| MUSE 7B (rank=16)     | 13.1M (0.19%)    | ≈104 min |
| KnowUndo 7B (rank=16) | 13.1M (0.19%)    | ≈55 min  |

Table 11: Trainable parameters and wall-clock time per BLADE run.

```
provides generic information not
overlapping with the target fact.
- 1 = Partial leakage. Some specific
  details overlap with the target (e.g.,
  correct name but wrong details).
- 2 = Full leakage. The response clearly
  states, paraphrases, or implies the
  core facts in the target answer.
```

```
### RESPONSE_QUALITY
- 0 = Incoherent. Gibberish, repetition
  loops, garbled tokens.
- 1 = Understandable but awkward,
  truncated, or repetitive.
- 2 = Fluent. Reads like natural,
  well-formed text.
```

```
## Inputs for this evaluation
**Question**: {question}
**Target fact**: {ground_truth}
**Model response**: {generation}
```

```
## Output format
{"forget_leakage": <0|1|2>,
 "response_quality": <0|1|2>}
```

Retain-set evaluation uses an analogous prompt replacing FORGET\_LEAKAGE with RETAIN\_ACCURACY (0=wrong, 1=partial, 2=correct). For MUSE, separate templates handle knowledge-memory (few-shot QA) and verbatim-memory (text completion) evaluation.

## B.5 Compute

All experiments run on a single NVIDIA H100 (95GB) Azure Standard\_NC40ads\_H100\_v5 instance with bf16 precision and gradient checkpointing.

**Total GPU budget.** Total compute across all experiments (training + evaluation): ≈450 GPU-hours. Breakdown: BLADE 5-fold (40h), baselines 5-fold (198h), finetuning (4h), ablations (94h), evaluation/generation (82h). See [results/gpu\\_budget\\_breakdown.md](#) for full itemization.

## B.6 Implementation Details

**LR calibration.** At 10% of the safety cap  $T$ , the observed  $\mathcal{L}_{\text{tgt}}$  decay rate is compared to a target pace. The outer LR is then rescaled

once, bounded to  $[0.3, 3.0] \times$  the initial value. This adapts bidirectionally without manual scheduling.

## C Extended Results

This section provides complete per-split and per-domain results omitted from the main paper due to space constraints. Table 12 reports all TOFU 3B metrics across three forget splits with 5-fold standard deviations; Table 13 gives the corresponding LLM judge scores; Table 14 reports KnowUndo results for both copyright and privacy domains; and Table 15 extends the main paper’s ablation (Table 7) with configuration IDs and additional failure modes.

**Key observations.** On TOFU 3B, BLADE and PDU are the only methods that consistently achieve near-zero forget leakage ( $\text{Prob} \leq 0.01$ ) across all splits while maintaining utility. BLADE wins on the LLM judge metric ( $\text{HM}_J \geq 0.962$ ) because it achieves lower forget leakage scores without sacrificing response quality. On KnowUndo, BLADE achieves the highest HM on both domains ( $+0.021$  over the next best on copyright,  $+0.059$  on privacy) and the highest judge HM ( $+0.06$  on copyright,  $+0.06$  on privacy), demonstrating that the bilevel constrained approach generalizes beyond synthetic forget sets to real-world copyright and PII removal.

## D Training Dynamics

Figure 3 shows BLADE dynamics across all benchmarks. The three-phase pattern is consistent: (1) *warm-up* (steps 1–20): forget loss drops rapidly while retain loss stays within  $\varepsilon$ ; (2) *spike-and-ratchet* (steps 20–80): transient retain violations trigger the dual ratchet, pushing  $\lambda$  up by discrete increments per Proposition 2; and (3) *smooth convergence* (remaining steps):  $\lambda$  stabilizes and both losses decrease monotonically. The final  $\lambda$  scales with task difficulty without any hyperparameter change:  $2.4$  (fgt01)  $\rightarrow 3.6$  (fgt05)  $\rightarrow 4.5$  (fgt10), reflecting the harder optimization landscape when more knowledge must be erased.

Figure 4 shows PDU’s contrasting behavior. The unbounded logit-margin loss and symmetric dual updates produce recurring retain oscillations that never converge: retain loss spikes by up to  $2 \times$  its baseline before recovering, with

the amplitude growing on larger forget sets. On fgt10, PDU’s retain loss exceeds  $4 \times$  baseline at step 150 before partial recovery. This instability explains PDU’s collapse at  $4 \times$  scale in the scalability experiment (Section 5.5).

Figure 5 shows the inner loop ablation:  $K=3$  averages  $+0.020$  HM over  $K=0$  across 8  $\varepsilon$ -multiplier settings. The benefit is most pronounced at tight constraints ( $\varepsilon_{\text{mul}} \leq 2.0$ ), where the inner loop provides the retain optimizer time to compensate before the outer loop takes another forget step. At loose constraints ( $\varepsilon_{\text{mul}} \geq 4.0$ ), the ALM penalty alone suffices and the inner loop contribution shrinks to  $<0.005$  HM.

## E Use of AI Assistants

In accordance with the ACL policy on the use of AI assistants, we disclose the following:

**Tool.** Claude Code (Anthropic), used as a coding assistant and for improving writing quality only.

**Scope of use.** Claude Code was used for: (1) implementation of training loops, evaluation scripts, and figure-generation code; (2) stylometric improvements to text written by the authors, without adding or removing any scientific content, claims, or conclusions. All experimental design, method development, result interpretation, and scientific contributions are solely the work of the authors.

**Verification.** All code outputs were reviewed and validated by the authors. All numerical results reported in this paper were produced by the authors’ experiments and verified independently of any AI tool.

Table 12: TOFU (Llama-3.2-3B), 5 seeds. Best HM in **bold** (excl. Gold).

| Split | Method         | MU $\uparrow$          | Prob $\downarrow$      | ROUGE $\downarrow$     | HM $\uparrow$          |
|-------|----------------|------------------------|------------------------|------------------------|------------------------|
| fgt01 | Gold (retrain) | .663                   | .179                   | .409                   | .656                   |
|       | GradAscent     | .668 $\pm$ .001        | .564 $\pm$ .009        | .525 $\pm$ .014        | .509 $\pm$ .008        |
|       | GradDiff       | .659 $\pm$ .003        | .569 $\pm$ .019        | .578 $\pm$ .027        | .483 $\pm$ .020        |
|       | NPO            | .668 $\pm$ .000        | .553 $\pm$ .009        | .494 $\pm$ .008        | .525 $\pm$ .005        |
|       | SimNPO         | .656 $\pm$ .002        | .920 $\pm$ .007        | .830 $\pm$ .021        | .151 $\pm$ .012        |
|       | RMU            | .657 $\pm$ .001        | .861 $\pm$ .002        | .715 $\pm$ .006        | .246 $\pm$ .003        |
|       | BLURNPO        | .667 $\pm$ .003        | .822 $\pm$ .015        | .788 $\pm$ .024        | .253 $\pm$ .021        |
|       | PDU            | <b>.692</b> $\pm$ .001 | .168 $\pm$ .006        | .283 $\pm$ .007        | .742 $\pm$ .003        |
|       | BLADE          | .654 $\pm$ .003        | <b>.000</b> $\pm$ .000 | <b>.015</b> $\pm$ .011 | <b>.846</b> $\pm$ .004 |
| fgt05 | Gold (retrain) | .659                   | .130                   | .388                   | .698                   |
|       | GradAscent     | .481 $\pm$ .015        | .140 $\pm$ .015        | .333 $\pm$ .013        | .632 $\pm$ .008        |
|       | GradDiff       | .566 $\pm$ .005        | .147 $\pm$ .008        | .395 $\pm$ .010        | .653 $\pm$ .003        |
|       | NPO            | .542 $\pm$ .005        | .219 $\pm$ .004        | .359 $\pm$ .012        | .640 $\pm$ .003        |
|       | SimNPO         | .657 $\pm$ .001        | .901 $\pm$ .002        | .818 $\pm$ .005        | .175 $\pm$ .004        |
|       | RMU            | .644 $\pm$ .001        | .602 $\pm$ .004        | .534 $\pm$ .003        | .483 $\pm$ .002        |
|       | BLURNPO        | .640 $\pm$ .022        | .683 $\pm$ .082        | .593 $\pm$ .091        | .412 $\pm$ .075        |
|       | PDU            | <b>.687</b> $\pm$ .001 | .009 $\pm$ .001        | .085 $\pm$ .009        | <b>.843</b> $\pm$ .002 |
|       | BLADE          | .655 $\pm$ .002        | <b>.007</b> $\pm$ .006 | <b>.028</b> $\pm$ .016 | .842 $\pm$ .005        |
| fgt10 | Gold (retrain) | .661                   | .115                   | .382                   | .698                   |
|       | GradAscent     | .000 $\pm$ .000        | .000 $\pm$ .000        | .003 $\pm$ .004        | .000 $\pm$ .000        |
|       | GradDiff       | .532 $\pm$ .015        | .079 $\pm$ .008        | .361 $\pm$ .024        | .662 $\pm$ .005        |
|       | NPO            | .549 $\pm$ .013        | .237 $\pm$ .005        | .353 $\pm$ .059        | .640 $\pm$ .015        |
|       | SimNPO         | .652 $\pm$ .001        | .889 $\pm$ .002        | .808 $\pm$ .004        | .191 $\pm$ .003        |
|       | RMU            | .647 $\pm$ .001        | .390 $\pm$ .010        | .472 $\pm$ .002        | .591 $\pm$ .004        |
|       | BLURNPO        | .385 $\pm$ .167        | .245 $\pm$ .163        | .314 $\pm$ .094        | .502 $\pm$ .099        |
|       | PDU            | <b>.680</b> $\pm$ .009 | <b>.000</b> $\pm$ .000 | <b>.029</b> $\pm$ .008 | <b>.857</b> $\pm$ .003 |
|       | BLADE          | .647 $\pm$ .004        | .005 $\pm$ .003        | .038 $\pm$ .014        | .836 $\pm$ .006        |

Table 13: LLM Judge on TOFU 3B (5 seeds, Claude Opus 4.7). Best HM<sub>J</sub> in **bold**.

| Split | Method     | FL $\downarrow$         | RA $\uparrow$    | rRQ $\uparrow$   | HM <sub>J</sub> $\uparrow$ |
|-------|------------|-------------------------|------------------|------------------|----------------------------|
| fgt01 | GradAscent | 1.335 $\pm$ .086        | 1.849 $\pm$ .003 | 1.998 $\pm$ .000 | .587 $\pm$ .045            |
|       | GradDiff   | 1.500 $\pm$ .066        | 1.818 $\pm$ .017 | 1.997 $\pm$ .001 | .490 $\pm$ .041            |
|       | NPO        | 1.230 $\pm$ .041        | 1.843 $\pm$ .004 | 1.998 $\pm$ .001 | .640 $\pm$ .019            |
|       | SimNPO     | 1.825 $\pm$ .053        | 1.839 $\pm$ .008 | 1.995 $\pm$ .001 | .220 $\pm$ .057            |
|       | RMU        | 1.585 $\pm$ .038        | 1.807 $\pm$ .002 | 1.995 $\pm$ .001 | .432 $\pm$ .027            |
|       | BLURNPO    | 1.730 $\pm$ .122        | 1.853 $\pm$ .003 | 1.998 $\pm$ .001 | .308 $\pm$ .109            |
|       | PDU        | 0.615 $\pm$ .042        | 1.722 $\pm$ .007 | 1.968 $\pm$ .004 | .828 $\pm$ .011            |
|       | BLADE      | <b>0.010</b> $\pm$ .014 | 1.845 $\pm$ .004 | 1.995 $\pm$ .001 | <b>.970</b> $\pm$ .002     |
| fgt05 | GradAscent | 0.693 $\pm$ .042        | 1.452 $\pm$ .021 | 1.981 $\pm$ .012 | .766 $\pm$ .009            |
|       | GradDiff   | 0.833 $\pm$ .007        | 1.293 $\pm$ .022 | 1.705 $\pm$ .062 | .677 $\pm$ .009            |
|       | NPO        | 0.738 $\pm$ .037        | 1.552 $\pm$ .009 | 1.995 $\pm$ .001 | .774 $\pm$ .009            |
|       | SimNPO     | 1.742 $\pm$ .013        | 1.854 $\pm$ .007 | 1.996 $\pm$ .001 | .305 $\pm$ .012            |
|       | RMU        | 1.107 $\pm$ .033        | 1.703 $\pm$ .008 | 1.991 $\pm$ .003 | .679 $\pm$ .014            |
|       | BLURNPO    | 1.397 $\pm$ .153        | 1.712 $\pm$ .054 | 1.997 $\pm$ .001 | .538 $\pm$ .078            |
|       | PDU        | 0.067 $\pm$ .013        | 1.749 $\pm$ .009 | 1.978 $\pm$ .003 | .941 $\pm$ .002            |
|       | BLADE      | <b>0.023</b> $\pm$ .026 | 1.817 $\pm$ .011 | 1.990 $\pm$ .004 | <b>.962</b> $\pm$ .004     |
| fgt10 | GradAscent | 0.000 $\pm$ .000        | 0.000 $\pm$ .000 | 0.000 $\pm$ .000 | .000 $\pm$ .000            |
|       | GradDiff   | 0.663 $\pm$ .029        | 1.164 $\pm$ .058 | 1.432 $\pm$ .199 | .648 $\pm$ .037            |
|       | NPO        | 0.613 $\pm$ .078        | 1.519 $\pm$ .042 | 1.992 $\pm$ .006 | .796 $\pm$ .011            |
|       | SimNPO     | 1.713 $\pm$ .006        | 1.858 $\pm$ .005 | 1.995 $\pm$ .002 | .331 $\pm$ .006            |
|       | RMU        | 0.851 $\pm$ .017        | 1.703 $\pm$ .006 | 1.989 $\pm$ .002 | .765 $\pm$ .005            |
|       | BLURNPO    | 0.726 $\pm$ .354        | 1.046 $\pm$ .508 | 1.508 $\pm$ .601 | .550 $\pm$ .161            |
|       | PDU        | 0.014 $\pm$ .005        | 1.726 $\pm$ .029 | 1.952 $\pm$ .038 | .940 $\pm$ .011            |
|       | BLADE      | 0.034 $\pm$ .028        | 1.837 $\pm$ .011 | 1.992 $\pm$ .003 | <b>.965</b> $\pm$ .004     |

Table 14: KnowUndo full results (5-fold). HM = hmean(1−fgt\_R, ret\_R, MMLU). Judge HM = hmean(1−FL/2, RA/2, rRQ/2).

| Method      | Copyright |             |             |             |             |             | LLM Judge   |             |             |            |
|-------------|-----------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------------|
|             | fgt_Acc↓  | fgt_R↓      | ret_Acc↑    | ret_R↑      | MMLU↑       | HM↑         | FL↓         | RA↑         | rRQ↑        | HM↑        |
| FT (target) | .854      | .244        | .894        | .336        | .439        | .456        | 1.50        | 1.80        | 1.13        | .44        |
| GradAscent  | .002      | .000        | .002        | .000        | .250        | .000        | 0.00        | 0.00        | 0.00        | .00        |
| GradDiff    | .002      | .000        | .642        | .199        | .386        | .348        | 0.00        | 0.81        | 0.53        | .41        |
| NPO         | .678      | .189        | .763        | .289        | .445        | .433        | 0.77        | 1.74        | 1.15        | .66        |
| SimNPO      | .303      | .048        | .788        | .317        | .435        | .461        | 0.05        | 1.73        | 1.12        | .75        |
| RMU         | .435      | .076        | .763        | .201        | .388        | .347        | 0.04        | 1.01        | 0.98        | .60        |
| PDU         | .045      | .011        | .836        | .285        | .442        | .442        | 0.01        | 1.59        | 0.93        | .68        |
| BLADE       | .332      | <b>.040</b> | <b>.856</b> | <b>.332</b> | <b>.449</b> | <b>.477</b> | <b>0.04</b> | <b>1.79</b> | <b>1.17</b> | <b>.78</b> |

| Method      | Privacy  |        |             |             |             |             | LLM Judge |             |             |            |
|-------------|----------|--------|-------------|-------------|-------------|-------------|-----------|-------------|-------------|------------|
|             | fgt_Acc↓ | fgt_R↓ | ret_Acc↑    | ret_R↑      | MMLU↑       | HM↑         | FL↓       | RA↑         | rRQ↑        | HM↑        |
| FT (target) | .940     | .665   | .949        | .698        | .445        | .450        | 1.82      | 1.70        | 1.95        | .23        |
| GradAscent  | .017     | .004   | .012        | .006        | .318        | .017        | 0.00      | 0.00        | 0.00        | .00        |
| GradDiff    | .049     | .039   | .708        | .406        | .428        | .513        | 0.04      | 0.92        | 1.23        | .62        |
| NPO         | .646     | .322   | .790        | .426        | .450        | .496        | 0.74      | 1.12        | 1.99        | .68        |
| SimNPO      | .516     | .302   | .889        | .557        | .438        | .544        | 0.58      | 1.42        | 1.88        | .77        |
| RMU         | .568     | .072   | .661        | .079        | .361        | .182        | 0.07      | 0.07        | 0.21        | .08        |
| PDU         | .055     | .022   | .813        | .452        | .444        | .546        | 0.04      | 1.09        | 1.50        | .72        |
| BLADE       | .306     | .128   | <b>.937</b> | <b>.608</b> | <b>.462</b> | <b>.605</b> | 0.27      | <b>1.46</b> | <b>1.87</b> | <b>.83</b> |

Table 15: Complete ablation on MUSE Books (seed=42). Each row changes one component from full BLADE (A0). A16 (ALM off): collapses at step 31. A15 (No LoRA): catastrophic retain collapse. A9 ( $\tau=1.0$ ): unclamped gradients cause 8.2% retain degradation. A18 (Swapped):  $\lambda$  explodes to 367. A5 (NPO): loss collapses without actual unlearning.

| ID  | Configuration                      | fk↓  | vm↓  | ret↑ | HM↑         | $\Delta$ ret | FL <sub>J</sub> ↓ | RA <sub>J</sub> ↑ | rRQ <sub>J</sub> ↑ | HM <sub>J</sub> ↑ |
|-----|------------------------------------|------|------|------|-------------|--------------|-------------------|-------------------|--------------------|-------------------|
| A0  | Full BLADE                         | .110 | .000 | .658 | <b>.823</b> | —            | 0.14              | 1.40              | 1.69               | .814              |
| A1  | $K=0$ (no inner loop)              | .072 | .000 | .631 | .819        | −.027        | 0.10              | 1.37              | 1.68               | .811              |
| A9  | $\tau=1.0$ (unclamped)             | .161 | .003 | .604 | .780        | −.054        | 0.20              | 1.34              | 1.64               | .785              |
| A15 | No LoRA (full FT)                  | .002 | .000 | .306 | .459        | −.352        | 0.00              | 0.70              | 1.25               | .550              |
| A16 | ALM off ( $\lambda=0$ , $\rho=0$ ) | .000 | .002 | .092 | .233        | −.566        | 0.01              | 0.24              | 0.28               | .117              |
| A18 | Swapped bilevel                    | .372 | .643 | .651 | .506        | −.007        | —                 | —                 | —                  | —                 |
| A4  | GA forget loss                     | .006 | .003 | .060 | .160        | −.598        | 0.01              | 0.10              | 0.18               | .093              |
| A5  | NPO forget loss                    | .457 | .997 | .662 | .009        | +.004        | 1.44              | 1.47              | 1.72               | .495              |
| A17 | Logit margin loss                  | .000 | .000 | .000 | .000        | −.658        | 0.00              | 0.00              | 0.00               | .000              |

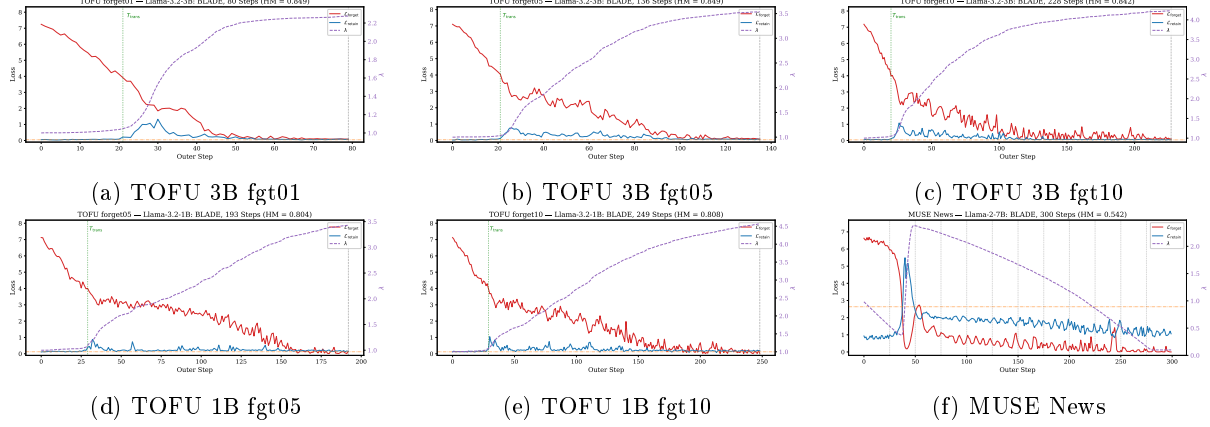


Figure 3: BLADE training dynamics across benchmarks. Three-phase pattern is consistent;  $\lambda$  adapts to task difficulty without hyperparameter change.

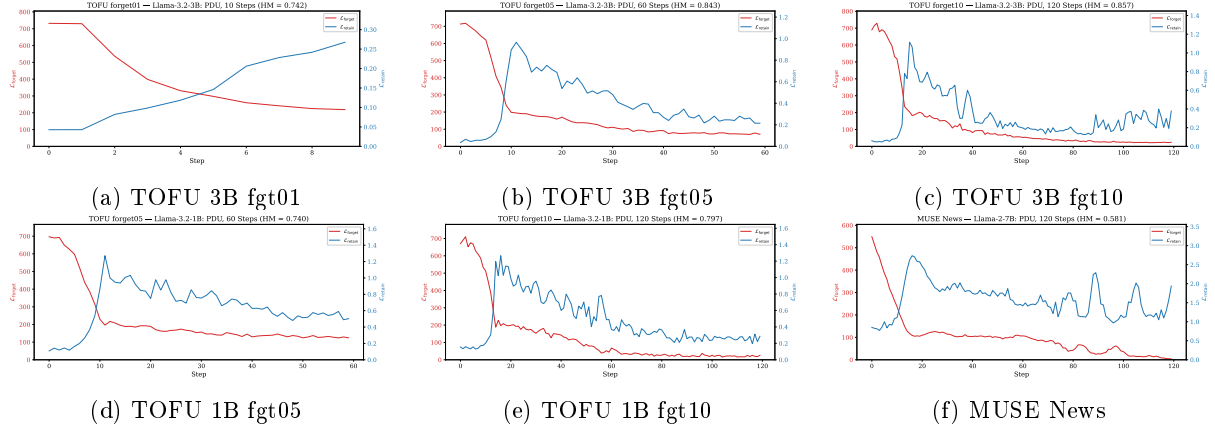


Figure 4: PDU training dynamics. Unbounded logit-margin loss and symmetric dual updates produce recurring retain oscillations across all benchmarks.

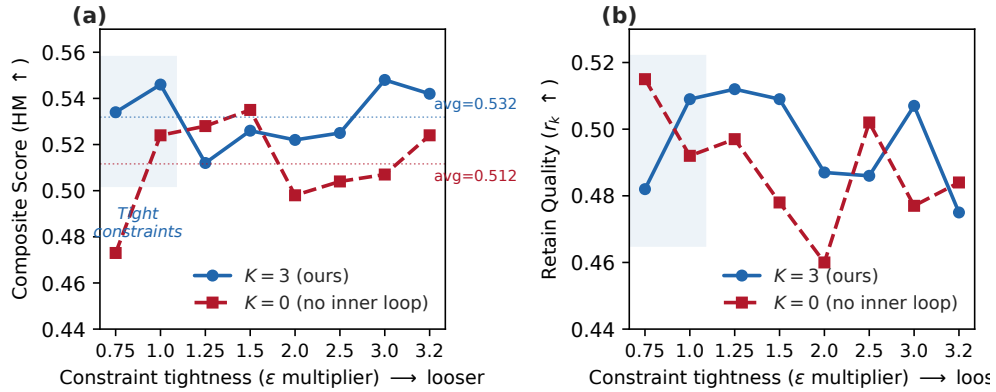


Figure 5: Inner loop ablation on MUSE News across 8  $\varepsilon$ -multiplier settings.  $K=3$  averages  $+0.020$  HM over  $K=0$ , with the gap widening at tight constraints.